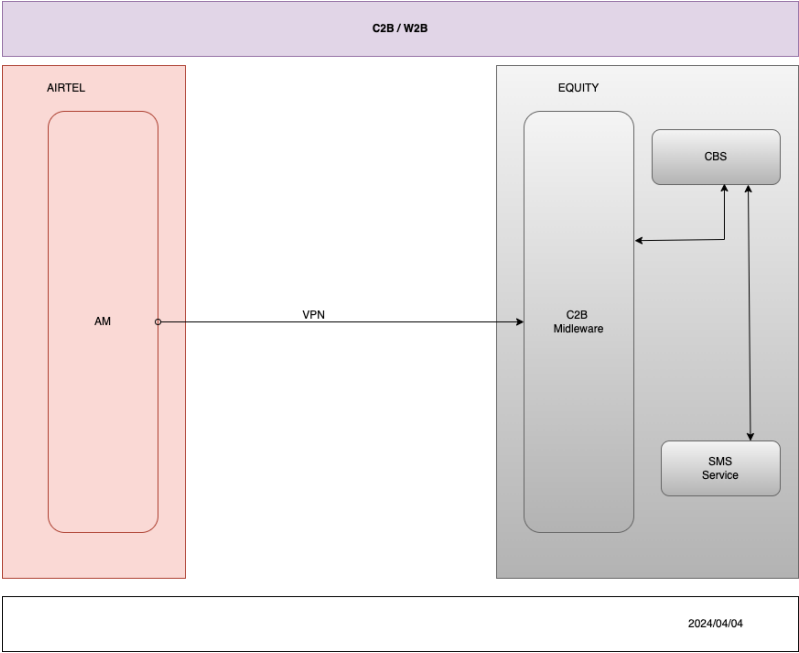


C2B API Specification

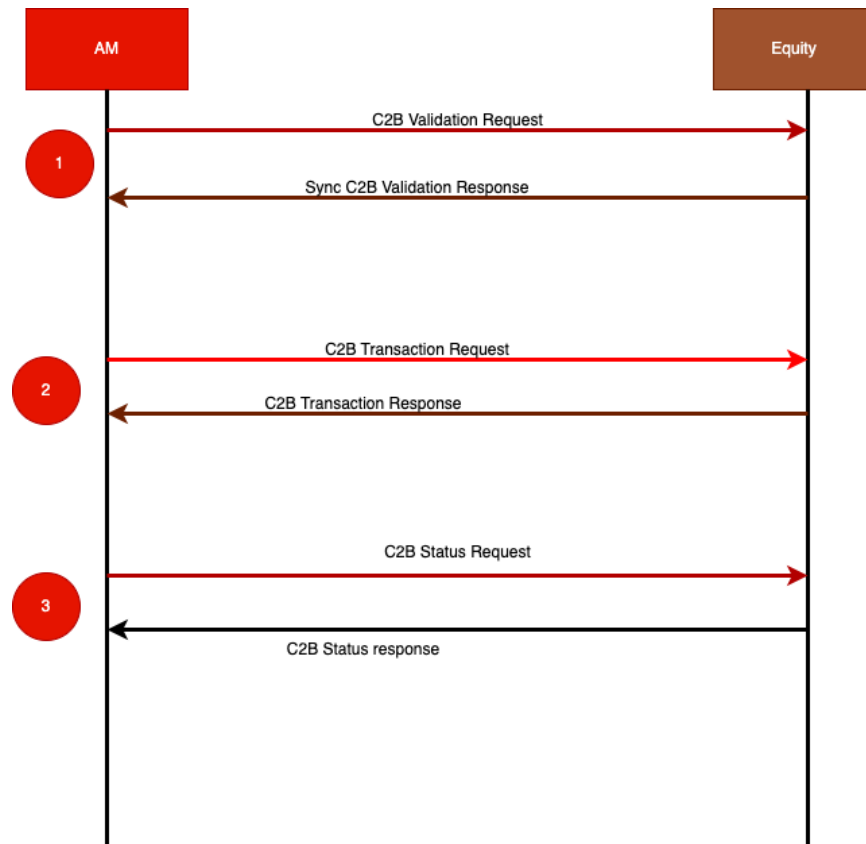
Introduction

C2B API interfaces Telecoms Wallet to Bank Service to Equity Banking System. It facilitates crediting of customer bank accounts and notification to the customer.

Overview diagram.



Transaction Message Sequence diagram:



Validation API

Request

Method: POST

Request: JSON

Endpoint: <https://196.216.242.163:8087/paybill/c2b/validate>

Validation Request Fields:

Field/Tag	Type	Remarks	Requirement
MessageType	String	VALIDATE TRANSACTION STATUS	Mandatory
transType	String	Transaction Type(SELF or OTHER)	Mandatory
transID	String	Wallet Transaction Id	Mandatory
transTime	String	Transaction timestamp	Mandatory

transAmount	String	Transaction Amount	Mandatory
businessShortCode	String	Pay-bill number or short code	Mandatory
billRefNumber	String	Customer reference	Mandatory
msisdn	String	Customer mobile Number: - The mobile number of the sender	Mandatory
customerName	String	Customer Names: - The name of the sender	Mandatory
countryCode	String	Two-character length ISO 3166 Alpha-2 Country code e.g TZ	Mandatory
currencyCode	String	Three character length ISO 4217 Currency code e.g TZS	Mandatory
signature	String	Concatenate TransType + TransID + TransTime + TransAmount + Msisdn Sign with private with SHA256withRSA algorithm and then base 64 encode	Mandatory

Validation Response Fields:

Field/Tag	Type	Remarks	Requirement
MessageType	String	VALIDATE TRANSACTION STATUS	Mandatory
transTime	String	Transaction timestamp	Mandatory
transAmount	String	Transaction Amount	Mandatory
businessShortCode	String	Pay-bill number or short code	Mandatory
responseCode	String	00 successful	Mandatory

responseDesc	String	successful	Mandatory
Account Name	String	The name of the account	Conditional
Account CurrencyCode	String	The currency of the account inquired	Conditional

Transaction API

Request

Method: POST

Request: JSON

Endpoint: <https://196.216.242.163:8087/paybill/c2b>

Transaction Request Fields:

Field/Tag	Type	Remarks	Requirement
transType	String	Transaction Type(SELF or OTHER)	Mandatory
transID	String	Wallet Transaction Id	Mandatory
transTime	String	Transaction timestamp	Mandatory
transAmount	String	Transaction Amount	Mandatory
businessShortCode	String	Pay-bill number or short code	Mandatory
billRefNumber	String	Customer reference	Mandatory
invoiceNumber	String	Customer reference or payment reference	Optional
orgAccountBalance	String	RRN or unique request IDPyabill Account Balance	Optional
msisdn	String	Customer mobile Number:- The mobile number of the sender	Mandatory

customerName	String	Customer Names:- The name of the sender	Mandatory
kycInfo	String	Customer other details	Optional
channelId	String	Source System	Mandatory
countryCode	String	Two-character length ISO 3166 Alpha-2 Country code e.g TZ	Mandatory
currencyCode	String	Three character length ISO 4217 Currency code e.g TZS	Mandatory
signature	String	Concatenate TransType + TransID + TransTime + TransAmount + BusinessShortCode + Msisdn Sign with private with SHA256withRSA algorithm and then base 64 encode	Mandatory

Transaction Response Fields:

Field/Tag	Type	Remarks	Requirement
transType	String	Transaction Type(SELF or OTHER)	Mandatory
transID	String	Wallet Transaction Id	Mandatory
transTime	String	Transaction timestamp	Mandatory
transAmount	String	Transaction Amount	Mandatory
businessShortCode	String	Pay-bill number or short code	Mandatory
billRefNumber	String	Customer reference	Optional
invoiceNumber	String	Customer reference or payment reference	Optional

orgAccountBalance	String	RRN or unique request IDPyabill Account Balance	Optional
msisdn	String	Customer mobile Number	Mandatory
customerName	String	Customer Names	Mandatory
kycInfo	String	Customer other details	Optional
channelId	String	Source System	Mandatory
rrn	String	transaction reference number	Mandatory
bankTransactionID	String	Transaction ID in Core Banking System	Optional
responseCode	String		Mandatory
responseDesc	String		Mandatory

Status API

Request

Method: POST

Request: JSON

Endpoint: <https://196.216.242.163:8087/paybill/c2b/status>

Status Request fields: -

Field/Tag	Type	Remarks	Requirement
MessageType	String	VALIDATE TRANSACTION STATUS	Mandatory
businessShortCode	String	Pay-bill number or short code	Mandatory
transID	String	Wallet Transaction Id	Mandatory
transAmount	String	Transaction Amount	Mandatory
countryCode	String	Two-character length ISO 3166 Alpha-2 Country code e.g TZ	Mandatory

currencyCode	String	Three character length ISO 4217 Currency code e.g TZS	Mandatory
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Status Response Fields:

Field/Tag	Type	Remarks	Requirement
transType	String	Transaction Type(SELF or OTHER)	Mandatory
transID	String	Wallet Transaction Id	Mandatory
transTime	String	Transaction timestamp	Mandatory
transAmount	String	Transaction Amount	Mandatory
businessShortCode	String	Pay-bill number or short code	Mandatory
billRefNumber	String	Customer reference	Optional
invoiceNumber	String	Customer reference or payment reference	Optional
orgAccountBalance	String	RRN or unique request IDPyabill Account Balance	Optional
msisdn	String	Customer mobile Number	Mandatory
customerName	String	Customer Names	Mandatory
kycInfo	String	Customer other details	Optional
channelId	String	Source System	Mandatory
rrn	String	transaction reference number	Mandatory
bankTransaction ID	String	Transaction ID in Core Banking System	Optional
responseCode	String		Mandatory
responseDesc	String		Mandatory

Sample Transaction request:

```
{
  "transType": "SELF",
  "transID": "GH123FGJ",
  "transTime": "",
  "transAmount": 300,
  "businessShortCode": "400200",
  "billRefNumber": "34NBH",
  "invoiceNumber": "",
  "orgAccountBalance": 2000000,
  "msisdn": "2547000000",
  "kycInfo": "John Doe",
  "signature": "dsdf2543534543csdffdfgfgf%$dfdfgfgf==",
  "channelId": "MGURUSH",
  "customerName": "John DOE",
  "countryCode": "TZ",
  "currencyCode": "TZS"
}
```

Sample Transaction response:

```
{
  "transType": "SELF",
  "transID": "GH123FGJ",
  "transTime": "",
  "transAmount": 300,
  "businessShortCode": "400200",
  "billRefNumber": "34NBH",
```



```
"invoiceNumber": "",
"orgAccountBalance": 2000000,
"msisdn": "2547000000",
"kycInfo": {"first_name": "John", "lastName":
            "Doe", "Id": "44545454554"},
"channelId": "MGURUSH",
"customerName": "John DOE",
"rrn": "2324343454",
"bankTransactionID": "DC3434",
"responseCode": "000",
"responseDesc": "posted Successful"
}
```

UAT End point

<https://196.216.242.163:8087/paybill/c2b>

Authentication

In the payload request signature is required. Signature is calculated as follows:

Concatenate TransType + TransID + TransTime + TransAmount + businessShortCode + msisdn

Sign with private with SHA256withRSA and base 64 encode

Sign the result text with private KEY and encode with Base64 , the result is placed in signature element

Error codes

Response

Error Code	Error Desc
0	Success
114	Failed Invalid Account
119	Failed General error
911	Timeout

-6	Validation failed signature check failed
-1	Validation failed Invalid message
-10	Failed Limit Error
77	Duplicate error